
The grid is complete.

Four arms, ten iterations, two graders —
and the answers interact.

Exp3 endpoint results · two paper drafts · Exp4 built and gated

Lior Baruch · Reichman University · 27 August 2026

Llama-3.2-1B therapist · gpt-4o-mini patient + training oracle · Claude Haiku 4.5 held-out judge · 96 personas · 8 instruments

All four arms complete — Exp3 is data-complete

Arm	Iterations trained	Model states scored (both graders)	GPU-hours
PTO K=0	1-10	Base + I1-I10 = 11	8.119
PTO K=5	1-10	Base + I1-I10 = 11	19.681
GRPO K=0	1-10	Base + I1-I10 = 11	27.906
GRPO K=5	1-10	Base + I1-I10 = 11	51.205

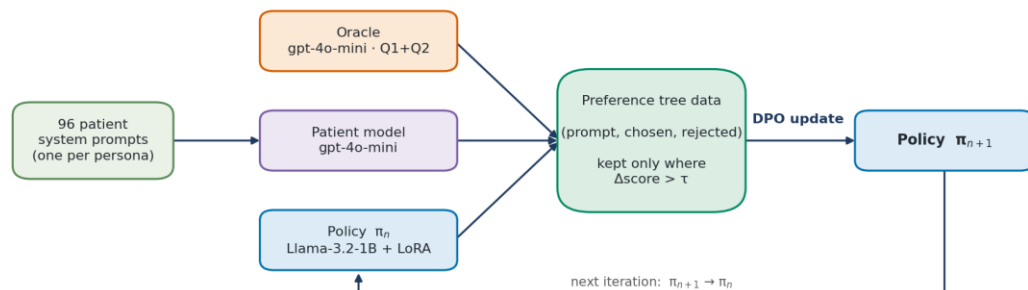
44 model states x 8 instruments x 96 personas = 33,792 scored cells per grader. Nothing is waiting to be scored; the matched iteration-10 endpoint every headline needed exists.

WHAT THIS MEANS

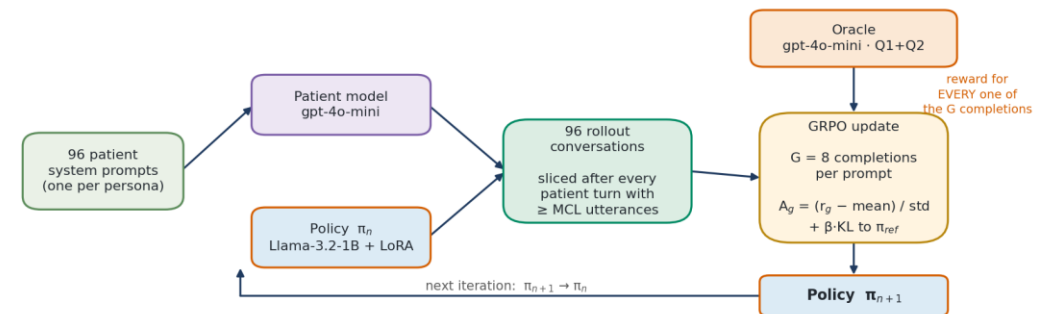
Training and scoring are finished. Everything from here is analysis and write-up — no further training spend in Exp3.

One 2x2: optimizer family x reward horizon

- **PTO vs GRPO.** Preference trees + DPO loss vs group-relative RL. Both iterative, sharing the same generation, oracle and look-ahead machinery; matched MCL=12, M = G = 8, temperatures, 96 patient personas per iteration.
- **K-turn look-ahead.** The oracle scores the K-turn continuation a candidate turn leads to, not the turn itself. K in {0, 5}, the same lever in both methods — only what the oracle sees changes, never the loss.
- **Reward vs eval.** Training reward is Q1+Q2 only (as in the ICLR paper). Evaluation is all 8 instruments: the 6 MI questionnaires + patient change-talk (PCT) + MI-inconsistency (MICI, lower = better).
- **Two graders.** gpt-4o-mini is the training oracle; Claude Haiku 4.5 is held out — a different model family that never played the patient and was never the reward. Levels are never compared across graders.



Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
 The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.



Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.
 As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

ACT I · EXP3 RESULTS

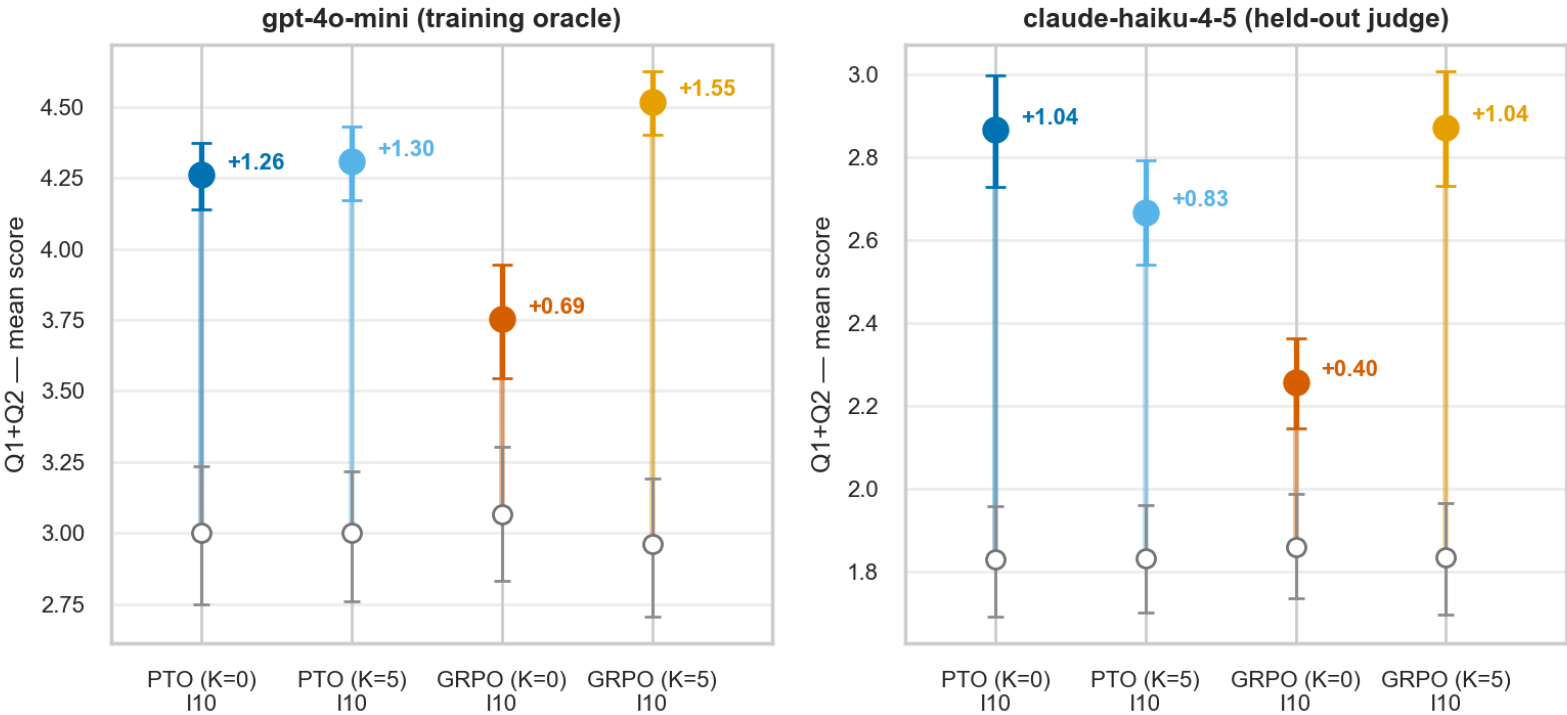
Four arms at the endpoint

Each research question now has an answer — and none of them is unconditional.

Where the four arms land — every instrument, both graders, each arm vs its own base

Q1+Q2 at the endpoint, one panel per grader. CIs = persona bootstrap, n = 96 personas. WARNING: INDEPENDENT y-axes - the panels are NOT on a common scale.

● endpoint (that arm's final iteration), 95% CI ○ that arm's OWN base (iteration 0), 95% CI — the gain (= delta in headline_grid)



WHAT Endpoint (iteration 10) change vs each arm's own base, per instrument, with persona-bootstrap CIs.	GRADERS Primary and held-out side by side on independent axes — levels are never compared across graders.	N 96 personas per model state; persona-paired throughout.	READ GRPO K=5 leads on the primary and holds on the held-out; GRPO K=0 is the weak arm on both.
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PTO vs GRPO: the verdict flips sign with K

PTO – GRPO, Q1+Q2 @ iteration 10	primary (gpt-4o-mini)	held-out (Claude Haiku 4.5)	verdict
K = 0	+0.507 (dz 0.729) ***	+0.609 (dz 1.265) ***	PTO wins
K = 5	-0.210 (dz -0.356) **	-0.206 (dz -0.313) *	GRPO wins

Sign: + = PTO higher. n = 96, persona-paired, Holm-corrected within grader. * p<.05 ** p<.01 *** p<.001.

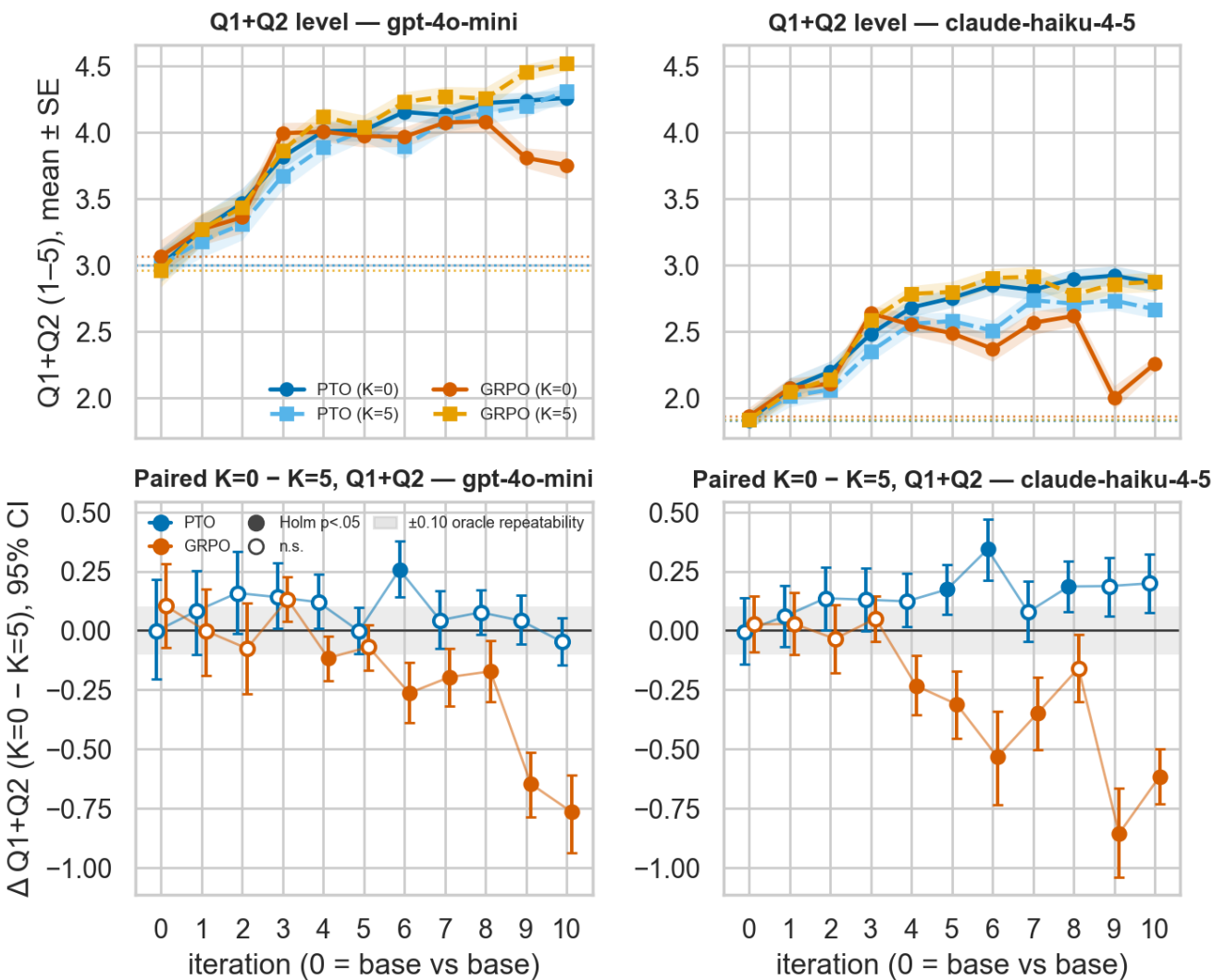
VERDICT

The method comparison is an interaction with the reward horizon: PTO wins at K=0, GRPO wins at K=5 — significant on both graders, at the same iteration, on the same conversations.

NOT THIS

“PTO beats GRPO” or “GRPO beats PTO”, unconditioned. Neither survives the table — a method claim without naming K is wrong at the other K.

The look-ahead lever by iteration — all four arms, both graders



SERIES
Q1+Q2 level curves for all four arms, base through iteration 10, plus the persona-paired K contrast strip.

GRADERS
Both, independent axes.

READ
The GRPO pair separates in K=5's favour from iteration ~4 and the gap is largest at the endpoint; the PTO pair never separates.

N
96 personas per state, persona-paired.

Look-ahead pays for GRPO — and not for PTO's reward

K=0 – K=5, Q1+Q2 @ iteration 10	primary	held-out	verdict
GRPO	-0.765 (dz -0.91) ***	-0.616 (dz -1.03) ***	K=5 clearly ahead
PTO	-0.047 (dz -0.10) n.s.	+0.199 (dz +0.31) n.s.	null

Sign: + = K=0 higher, so negative = K=5 ahead. GRPO's K=5 advantage is Holm-significant on both graders at iterations 4, 6, 7, 9 and 10 — largest at the endpoint. PTO never significantly favours K=5 at any iteration on either grader.

THE INTERACTION ITSELF

Difference-in-differences at iteration 10: 0.718 (dz 0.793) on the primary, 0.815 (dz 0.972) on the held-out judge — both $p_{\text{holm}} < .001$. The training grader is not blind to it, and neither is the grader that was never the reward.

GRPO K=5 @10 is the best final state on both graders

Final state @ 10	Q1+Q2 primary	Q1+Q2 held-out	MICI ↓ primary
GRPO (K=5)	4.517	2.873	0.210
PTO (K=5)	4.307	2.667	0.264
PTO (K=0)	4.260	2.866	0.491
GRPO (K=0)	3.753	2.257	0.838

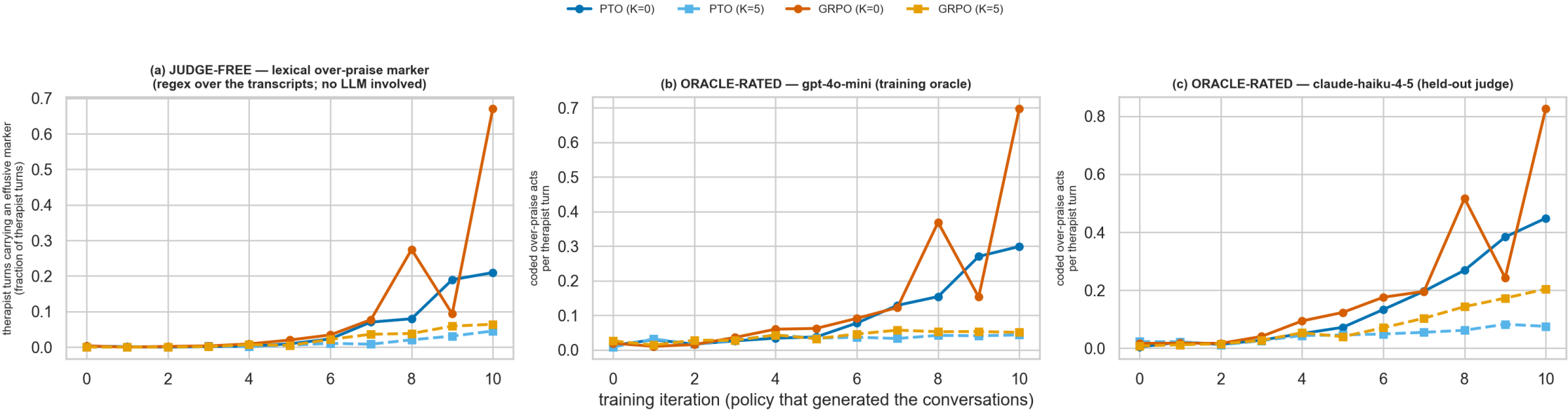
On the primary, GRPO K=5 leads every one of the 8 instruments at the endpoint and has the lowest MI-inconsistency of any final state.

READ IT CAREFULLY

“Still climbing” is a primary-grader statement: on the held-out judge the arm is flat since ~iteration 6 ($I_{10} - I_7 = -0.039$, $p = .65$) and statistically tied with PTO K=0 at the top. And the measurement caveat, three slides on, travels with 4.517.

The reward hack, measured without asking any judge

Over-praise: the judge-free measurement beside the two graded ones (K=0 solid · K=5 dashed)



WHAT A deterministic lexical over-praise marker computed from the transcripts — byte-identical under any grader — beside both graders' rated over-praise rates.	AXIS Share of therapist turns containing at least one marker (not a per-turn count).	READ Both K=0 arms climb steeply in the late iterations; both K=5 arms stay near zero.	N 96 conversations per state.
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Look-ahead suppresses over-praise in both optimizers

Arm @ 10	lexical marker rate (judge-free)	rated over-praise rate (primary)
GRPO (K=0)	0.671	0.698
PTO (K=0)	0.210	0.299
GRPO (K=5)	0.064	0.051
PTO (K=5)	0.045	0.043

VERDICT

0.671 → 0.064 within GRPO (10.5x) and 0.210 → 0.045 within PTO (4.7x). The marker is computed from the transcripts, so no grader dispute applies — and PTO K=0, not GRPO K=5, is the second-most-sycophantic arm.

NOT THIS

“K=5 stops the hack.” On the held-out grader, GRPO K=5's total MI-inconsistency still rises (0.326 → 0.628), at roughly half of K=0's rise; the primary reads the same conversations as flat. Say “slows the loop ~10x on over-praise”, per grader — never “stops”.

What K=5 buys each optimizer

GRPO

Reward. Its best endpoint on both graders: +0.765 / +0.616 over K=0 (dz ~0.9 / 1.0), ahead on all 8 instruments under the primary.

Reward hacking. Over-praise capture 10.5x lower (0.671 → 0.064, judge-free).

PTO

Reward. Nothing. No significant K=5 gain at any of 10 iterations on either grader; the endpoint contrast is null.

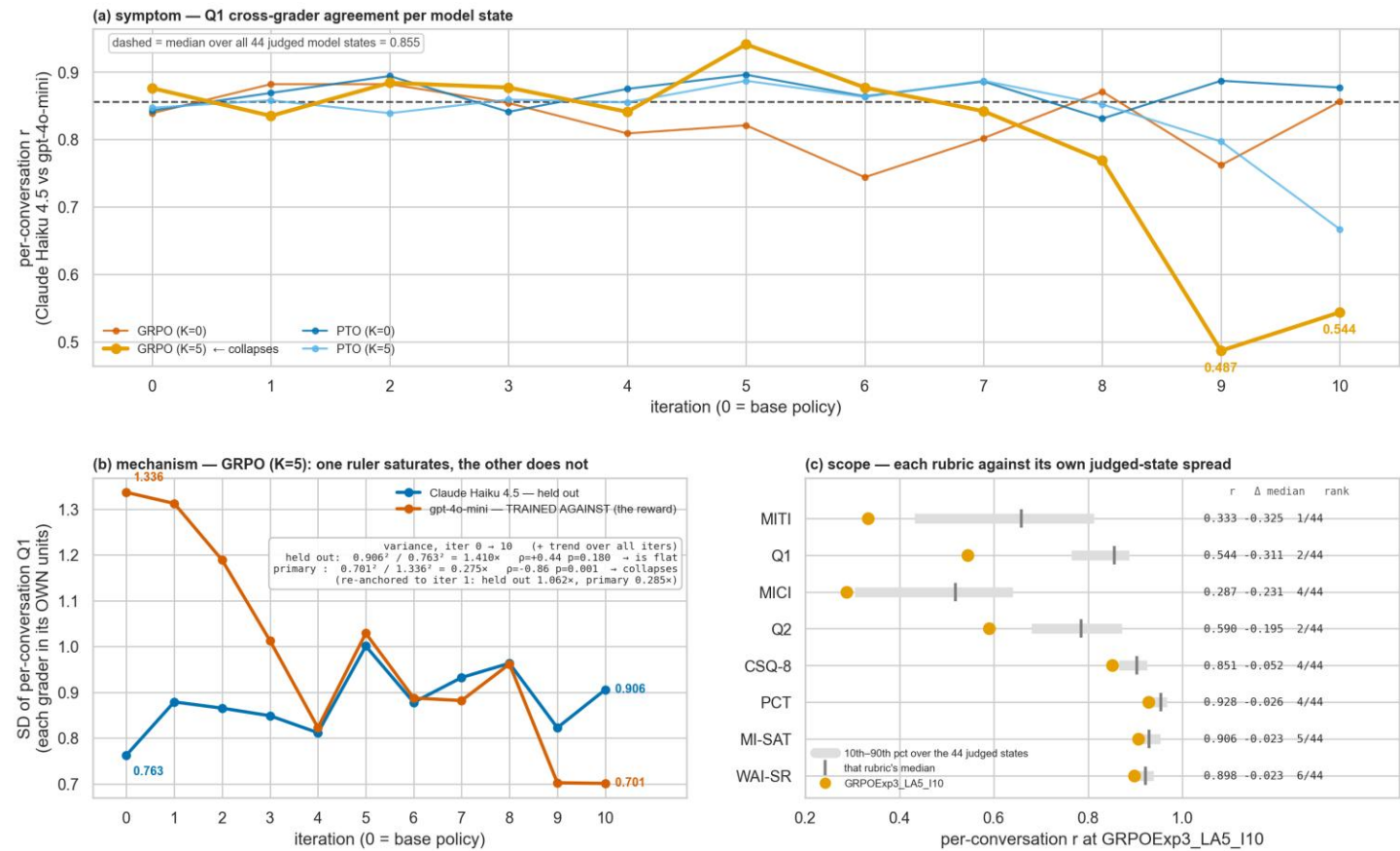
Reward hacking. Over-praise capture 4.7x lower (0.210 → 0.045, judge-free) — the lever still buys integrity where it buys no reward.

SAME LEVER, DIFFERENT OPTIMIZER

For GRPO, look-ahead is a performance lever. For PTO, it is only a hygiene lever. That asymmetry — not either method's win — is the finding of the experiment.

The caveat that travels with 4.517 — the training grader saturates on the winning arm

[EVAL] Judge saturation — Q1 cross-grader agreement collapses over GRPO (K=5)'s late iterations, because the TRAINED-AGAINST grader's spread collapses while the held-out grader's is flat



AGREEMENT

Per-conversation cross-judge agreement on Q1 along GRPO K=5: 0.941 @I5 → 0.487 @I9 → 0.544 @I10 — the two lowest of all 44 states (median 0.855).

MECHANISM

One-sided: the primary's Q1 SD collapses monotonically to 0.275x of base (rho -0.86, p=.001); the held-out judge's SD on the SAME conversations does not move.

SURVIVES

The arm-level ranking: the held-out judge independently puts K=5 ahead at dz 1.03. What breaks is the per-conversation ruler, exactly where the primary scores highest.

RULE

Never present the 4.517 endpoint without this slide.

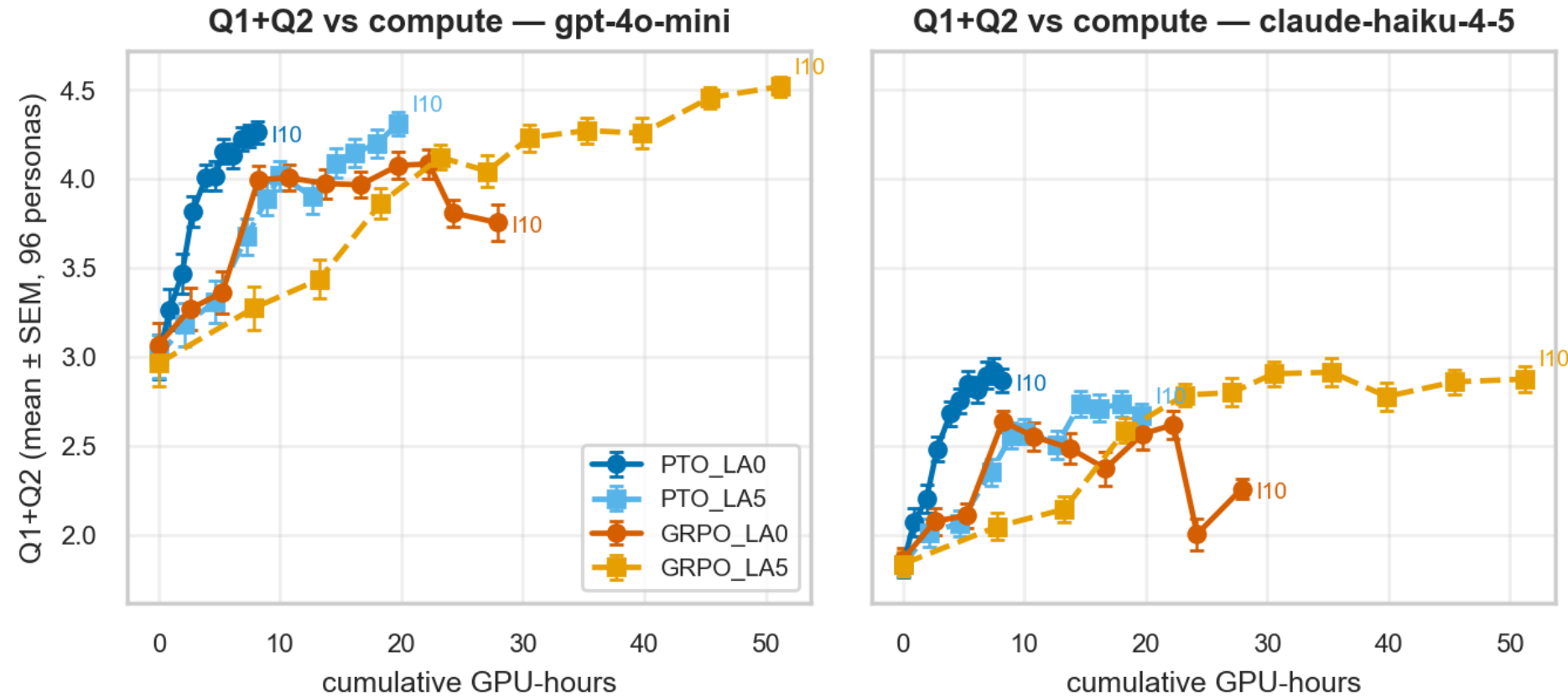
The replicate draw: every headline survives

Contrast, Q1+Q2	primary orig → replicate	held-out orig → replicate
K lever @10 (GRPO K5 – K0)	+0.765 → +0.709 *	+0.616 → +0.637 *
method @K0 (PTO – GRPO)	+0.507 → +0.516 *	+0.609 → +0.659 *
method @K5 (PTO – GRPO)	−0.210 → −0.155 *	−0.206 → −0.227 *
top pair (GRPO K5 – PTO K0)	+0.257 → +0.193 *	+0.007 → −0.022 (n.s. both)

Replicate = a fresh 96-conversation generation from the same two adapters (GRPO K5 @10, PTO K0 @10), scored on all 8 instruments by both graders. * = Holm-significant in both draws. The K-lever and method rows re-draw one side (the contested arm).

AND A NOISE FLOOR, FOR THE FIRST TIME
36 same-policy contrasts (2 arms x 9 metrics x 2 graders): 0 significant after Holm, max |dz| 0.216. The 4.517 headline reads 4.461 on the replicate; the held-out “tie” between GRPO K5 and PTO K0 is now demonstrated across two draws, not inferred from one gap.

The same curves on the axis money moves on — GPU-hours



X-AXIS Cumulative GPU-hours per arm, reconstructed from artifact mtimes (never from the per-process metadata, which undercounts resumed iterations).	COSTS PTO K0 8.119 h · PTO K5 19.681 h · GRPO K0 27.906 h · GRPO K5 51.205 h. Look-ahead costs 1.84x within GRPO; GRPO K5 costs 6.31x PTO K0.	READ An iteration is not a unit of spend — a whole PTO iteration costs a fraction of a GRPO one, so matched-iteration and matched-budget answer different questions.
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At matched budget the verdict changes

- **GRPO's K at 51.2 GPU-h:** $K=5 > K=0$ under all four grader select/evaluate combinations (Δ 0.256–0.435, every $p_{\text{holm}} < .001$), honest cross-grader selection included. The lever first draws level at ~23 GPU-h — below that it loses.
- **PTO's K at 19.7 GPU-h:** $K=5 < K=0$ or no significant difference on all four combinations — the opposite sign, so the interaction holds at matched budget as well as matched iteration.
- **Method at PTO's budgets:** PTO $\gg$ GRPO. At 8.1 GPU-h: 4/4 combinations (Δ 0.76–0.90, dz 1.07–1.39). At 19.7: 3/4 — because at PTO's whole-run budget GRPO $K=5$ has only reached iteration 3.
- **The sharpest sentence:** at 23.2 GPU-h GRPO $K=5$ is at iteration 4, scoring 4.120 / 2.784 — while PTO $K=0$'s entire ten-iteration run costs 8.119 GPU-h and scores 4.260 / 2.866: higher on both graders at 35% of the compute.

BOTH AXES, ALWAYS

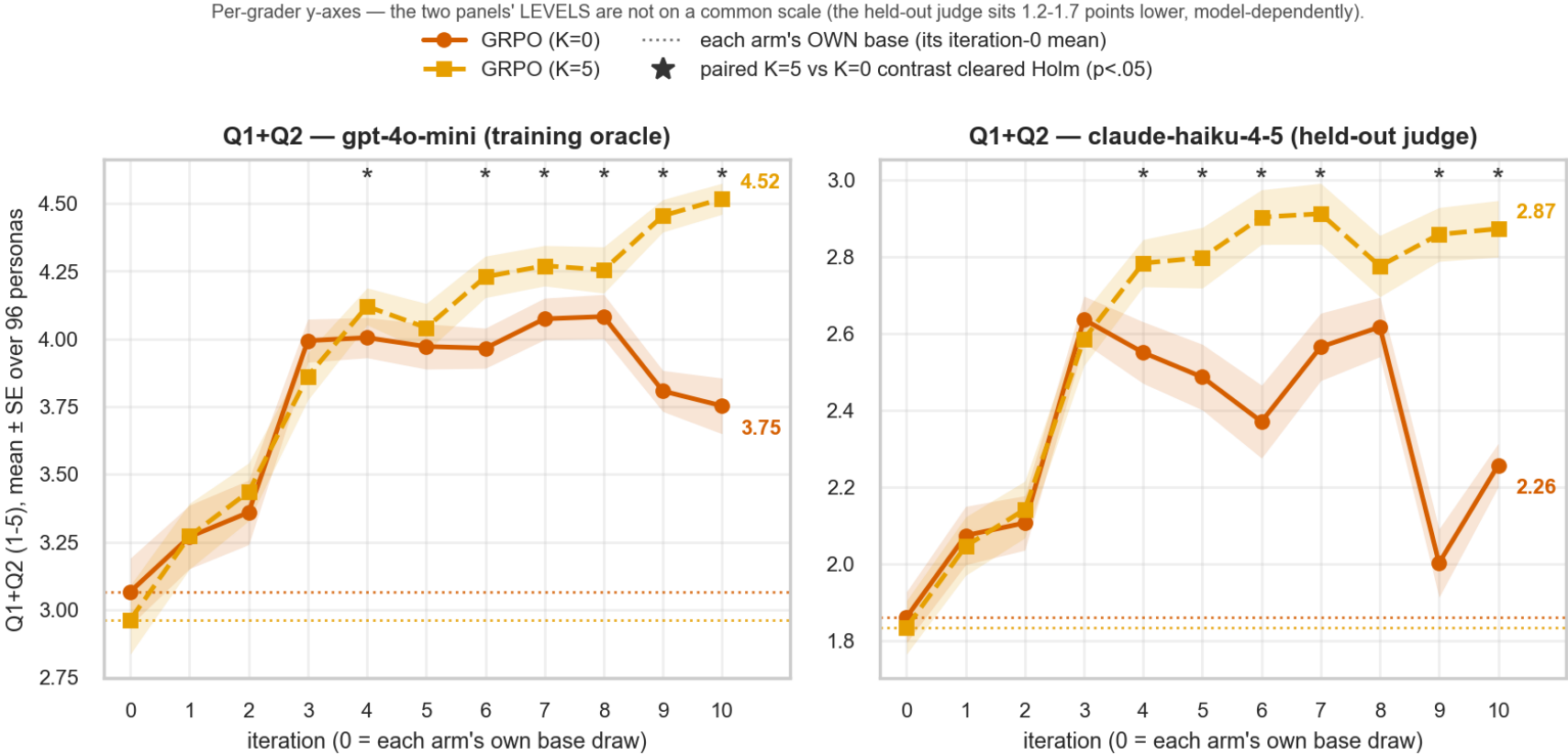
At matched iteration GRPO $K=5$ wins; at any budget PTO can afford, PTO wins. Neither statement replaces the other — quote the axis with the claim.

ACT II · PAPERS

Two drafts, disjoint by design

P1 takes the narrowest decisive question first; P2 owns everything four-arm.

Scoring the Continuation — K-turn look-ahead rewards for GRPO



ARGUMENT	SCOPE	STATUS	DEADLINES
Scoring a candidate turn by the K-turn continuation it leads to more than doubles what group-relative RL extracts from the same oracle — and is the difference between learning MI and learning to flatter the judge	The two GRPO arms ONLY. PTO appears nowhere — deliberately: every full-grid statistic was recomputed on the 22 GRPO states, and the paper defers optimizer generality to companion work.	Drafted end to end: 9-page ICLR body met, clean build, no TODOs, complete NUMBERS.md ledger. Double-blind until camera-ready.	Abstract 18 Sep (3 weeks) · full paper 25 Sep 2026, both 23:59 UTC-12.

Five moves — and what the draft refuses to claim

THE MOVES

- **§4 Reward.** K=5 beats K=0 on all 8 instruments under both graders; endpoint +0.765 / +0.616, gain vs base 2.3–2.6x K=0's.
- **§5 Cost, honest.** 1.92x per step, 1.835x per run; the lever loses below ~18 GPU-h and wins at the full budget under all four grader combinations.
- **§6 Behaviour.** The 10.5x judge-free over-praise result. Stated per grader: “slows and roughly halves the loop”, never “stops”.
- **§7 Mechanism.** Consistent-with, not shown: the K=5 training cut is a slightly better rank proxy; look-ahead rescales rather than sharpens the group signal.
- **§8 Measurement.** The saturation caveat travels with the headline, in the paper itself.

THE REFUSALS

- Not a claim about optimizers in general — PTO is never named; transfer is deferred to the companion paper.
- Not “eliminates reward hacking” — the per-grader split is stated.
- No dose-response: K in {0, 5} by design, and the limitations say so.
- No clinical claim — every instrument is LLM-administered; no human MI coder has rated any conversation.
- No patient-generalisation claim — all 96 personas are in-sample.

Same Lever, Different Optimizer — the full 2x2

- **The interaction.** The optimizer ranking flips with the reward horizon — DiD dz 0.79 / 0.97, both graders, every cell surviving the replicate draw.
- **The budget axis.** PTO K=0 matches GRPO K=5's held-out gain at 6.31x less compute; the reversals between matched-iteration and matched-budget are part of the claim.
- **The behaviour story.** Both K=0 arms flatter; look-ahead suppresses over-praise in both optimizers — this is where PTO's “hygiene lever” result lives, the arm of the story P1 deliberately leaves out.
- **Also inside.** The ICLR-era regime re-scored under the modern grader, and the full-grid measurement section (sign preservation 88.4% over 7,568 contrasts).
- **Status.** Drafted 2026-08-26 in ACL long format, 8-page body met, fresh NUMBERS.md. Target: the ARR October 2026 cycle, feeding NAACL 2027 + COLING 2027.

DISJOINT SCOPES

P1 never names PTO; P2 owns everything four-arm. Shared numbers are cited from the same EDA artifacts in both ledgers, and neither paper imports the other's argument.

ACT III · EXP4

The cost ceiling, removed

The same science on a fully open model stack — an arm costs \$0 in API.

Exp3's binding constraint was the API bill

- **What happened.** The OpenAI balance hit zero on 24 Aug, killing GRPO K=5's iteration 10 at step 70/136; the 20 Aug spend cap had already killed two sessions. Cost — not GPU — capped every design decision in Exp3 (iterations, G/M, the held RQ-iii).
- **The move.** Same science — same 1B therapist, same 96 personas and rubrics verbatim — but patient, oracle and judge are an open Gemma-4 model (E4B default) behind one local vLLM server. A full arm costs \$0 in API; only Colab GPU-hours.
- **The guard.** The grader IS the instrument, so an oracle-sanity gate blocks any arm until the Gemma grader proves non-degenerate spread and rank agreement against frozen gpt-4o-mini reference scores. An open grader can honour the schema and still measure nothing — that failure is silent, hence the gate.
- **Status.** Code complete and locally gated: 94 smoke checks + the EDA self-check green, real DPO and GRPO steps run on the local card. Nothing trained yet; no data exists.
- **By construction.** Five Exp3 defects fixed from day 1: per-attempt API timeouts, stable persona-id filenames, append-only per-phase timing, a per-state parquet score lake, append-only run-metadata history.

SCOPE

A side project — not a thesis chapter unless the results earn it. Different grader = different score axis: Exp4 numbers are never comparable to Exp3's.

The experiments Exp3 could not afford

- **RQ-iii, finally.** The oracle-instrument sweep — train on WAI-SR, CSQ-8 or MITI instead of Q1+Q2. Held in Exp3 purely for cost; the training questionnaire is a first-class knob in Exp4's arm grammar.
- **K as a dose.** K in {1, 2, 3, ...} between the two endpoints — the dose-response curve P1 explicitly refuses to claim on two points.
- **Training-run replicates.** Exp3 has ONE training run per arm; the replicate draw bounds evaluation noise only. At \$0 API, seed replicates of whole arms become affordable.
- **Longer runs.** GRPO K=5 was still climbing on the primary at iteration 10 — the truncation was budget, not convergence.
- **The gate ladder.** Colab roles smoke → full oracle-sanity on both Gemma sizes (choose the grader by rank agreement + spread) → 2-iteration mini-arm → first real arm (GRPO K=0).

THE ASK

One Colab session to run the gate ladder end to end — GPU-hours only, \$0 API. If the Gemma grader passes sanity, the first real arm can start immediately after.

Three decisions, one calendar

	What	When	The decision
1	P1 — Scoring the Continuation	ICLR 2027: abstract 18 Sep (3 weeks), paper 25 Sep	Freeze scope now; co-author list + final read-through; remaining items are its README's open list.
2	P2 — Same Lever, Different Optimizer	ARR October 2026 cycle	Polish after P1 freezes; the drafts share numbers only through the EDA artifacts.
3	Exp4 gate ladder	one Colab session, \$0 API	Go / no-go: run now in the background of writing, or hold until after the P1 deadline.

SUGGESTED ORDER

P1 first — it is the nearest deadline and needs no new data. The Exp4 gates are a single session and can run while writing; any real Exp4 arm waits until P1 is submitted.

Look-ahead is a performance lever for GRPO **and a hygiene lever for PTO.**

Which optimizer wins depends on which K — and which axis — you ask.

The grid that settles this is complete, replicated at its contested corners, and priced.

Next: P1 to ICLR 2027 (abstract in 3 weeks) · P2 to ARR October · the Exp4 gate ladder, one Colab session.